Toward sustainable and resilient fisheries management for the Humboldt squid deep-water fishery

emLab, UCSB

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Introduction

Here we describe the data sources and exploratory data analysis for the project, **Toward sustainable and resilient fisheries management for the Humboldt squid deep-water fishery**. The project is a collaboration between UCSB's Environmental Markets Lab (emLab) and Environmental Defense Fund (EDF).

Use of generative AI

Claude (Anthropic), model Opus 5.5, was used as a coding assistant through Claude Code in November 2026. It helped move the AIS data to Global Fishing Watch's V5 pipeline, extend the data through 2025, split the data pipeline into separate projects for the GFW, SST, and analysis steps, switch the SST download to NOAA's NCEI archive, read the CMIP6 forecasts from the IPCC Atlas dataset, and revise the text and figures in this book.

1 Dataset descriptions

1.1 Sea surface temperature (SST)

Sea surface temperature (SST) data come from NOAA's daily Optimum Interpolation Sea Surface Temperature (OISST) version 2.1 (Huang et al. (2021)). We download the final (not preliminary) daily files from NOAA's National Centers for Environmental Information (NCEI) archive, and check that each file holds the expected date before using it. The raw data are provided globally at 0.25x0.25 degree daily resolution. We keep the pixels within the geographic analysis scope (see below), and temporally aggregate the daily data to monthly resolution by calculating the mean (and standard deviation, minimum, and maximum) SST for each pixel from across the days in each month. Every day from 2016-01-01 through 2025-12-31 is included.

1.2 Oceanic Niño Index (ONI)

Monthly Oceanic Niño Index (ONI) values come from NOAA's Physical Sciences Laboratory. The ONI is NOAA's primary index of the El Niño-Southern Oscillation (ENSO): the 3-month running mean of sea surface temperature anomalies in the Niño 3.4 region (5°N-5°S, 120°-170°W), from NOAA's Extended Reconstructed SST (ERSST) dataset. Values of +0.5°C or higher indicate El Niño conditions, and -0.5°C or lower indicate La Niña conditions. NOAA revises past ONI values when it updates ERSST or the index's base periods, so values for earlier months can shift slightly between data pulls. To keep the analysis reproducible, we use a copy of the ONI file saved on 2026-11-02, and use the ONI for each month from 2016-01-01 through 2025-12-31.

1.3 Sea surface temperature (SST) forecasts under climate change

We use SST forecasts under climate change from the IPCC WGI Interactive Atlas(Iturbide et al. (2021)). The Atlas provides a platform for accessing ensemble forecasts from CMIP6 (Coupled Model Intercomparison Project Phase 6), which represent the latest global climate forecasts available and served as the basis of the 6th IPCC Assessment Report.

We use the data underlying the Atlas, the IPCC-WGI AR6 Interactive Atlas Dataset (Instituto de Física de Cantabria, Version 2 final), which has monthly SST from each CMIP6 model interpolated to a common 1x1 degree grid. We read it from the dataset's THREDDS server, for the 1x1 degree cells within the geographic analysis scope. For each scenario and time horizon, we calculate each model's mean SST for each calendar month across the years of the time horizon it has data for, then average across the models with SST in each cell. This follows the Atlas's own code (`datasets-interactive-atlas/04_map_figures.R` in Iturbide et al. (2021)), and reproduces the values on the Interactive Atlas to within 0.02°C. A few model runs don't cover every year (e.g., the CAMS-CSM1-0 and IITM-ESM runs end before 2100, and EC-Earth3-Veg-LR is listed for SSP5-8.5 but has no SST data, so SSP5-8.5 means use 26 models); the models and years used are listed in `cmip6_models_used.csv` in the pipeline outputs. We use three time horizons:

1. Near Term (2021-2040)
2. Medium Term (2041-2060)
3. Long Term (2081-2100)

And for each time horizon, four climate change scenarios (more information on the scenarios can be found here), each with the models the Atlas uses for it:

1. SSP1-2.6 (26 models)
2. SSP2-4.5 (28 models)
3. SSP3-7.0 (24 models)
4. SSP5-8.5 (27 models)

This gives us a total of 12 different forecasts to analyze. The pipeline checks that each has all 12 months, and that no two are identical. (Until November 2026, we downloaded these forecasts by hand from the Interactive Atlas website. Some of those files turned out to be copies of a different scenario or time horizon, and the Near Term files had no February.) The dataset can be loaded in targets using `targets::tar_load(sst_cc_forecast_data)`, and it has the following columns:

- **lon_bin**: 1x1 degree longitude bin, lower-left corner (degrees) (numeric)
- **lat_bin**: 1x1 degree latitude bin, lower-left corner (degrees) (numeric)
- **sst_deg_c_mean**: Multi-model mean sea surface temperature for the calendar month (degrees C) (numeric)
- **time_period**: Future time horizon forecast time period (character)
- **scenario**: Climate change scenario (character)
- **month_number**: Month number (numeric)

1.4 AIS-based fishing effort

We use satellite-based individual vessel monitoring AIS data processed by Global Fishing Watch (Kroodsma et al. (2018)). We use the V5 pipeline table `global-fishing-watch.pipe_ais_v5_published.messages`. Variables of interest within this table include the following (descriptions are taken directly from the schema for `pipe_ais_v5_published.messages`):

- `ssvid`: source specific vessel id; MMSI for AIS
- `hours`: time since the previous position in the segment
- `timestamp`: timestamp for position
- `lon`: longitude
- `lat`: latitude
- `night_loitering`: 1 if the seg_id of every message of a squid_jigger that is at night and not moving, 0 if not.

In order to minimize noisy data, we only include AIS messages that have a `clean_segs` boolean (i.e., all messages must have `good_seg` boolean and must not have an `overlapping_and_short` boolean). We filter to just those messages where `night_loitering = 1`. For squid jigging vessels, GFW uses the heuristic of night loitering to identify when they are fishing. Therefore, any `hours` where `night_loitering = 1` can be classified as `fishing_hours`

We take the raw high-resolution AIS data and aggregate `fishing_hours` spatially (by 0.25x0.25 degree pixels, which are roughly 27.75km x 27.75km at the equator), temporally by month, and by flag. We currently process data from 2016-01-01 through 2025-12-31.

1.5 AIS-based vessel info

Vessel characteristics data processed are by Global Fishing Watch (Park et al. (2023)). We use the V5 pipeline table `global-fishing-watch.pipe_ais_identity_v5_partners_published.vi_ssvid`, which has one row per vessel summarizing its whole AIS record. In V5, vessel class, whether a vessel is a fishing vessel, and engine power come from GFW's random forest vessel classification model (`random_forest_info`), which replaces the V3 fields `best.best_vessel_class` and `on_fishing_list_best` (GFW Pipe5 Release Notes for Research Partners, September 2026). Variables of interest within this table include the following (descriptions of the `random_forest_info` fields are from the release notes, since the schema does not describe them; the others are taken directly from the schema):

- `ssvid`: source specific vessel id; MMSI for AIS
- `best.best_flag`: best flag state (ISO3) for the vessel
- `random_forest_info.on_fishing_list_best`: whether the vessel is a fishing vessel, from the random forest model, overridden by official registry information where available

- `random_forest_info.best_vessel_class`: best vessel class for the vessel from the random forest model. The model only names a specific gear when it scores that gear at least 0.35; otherwise it rolls the vessel up to a broader class, such as the generic `fishing`
- `random_forest_info.atomic_class`: the random forest model's finest-level vessel class prediction
- `random_forest_info.best_engine_power_kw`: best engine power (kilowatts) for the vessel (using official registry information where available, or the random forest model's estimate where not available)
- `activity.active_hours`: hours the vessel was broadcasting AIS and moving more than 0.1 knots
- `activity.offsetting`: true if this vessel has been seen with an offset postition at some point between 2012 and 2019
- `activity.overlap_hours_multinames`: the total numbers of hours of overlap between two segments where, over the time period of the two segments that overlap (including the non-overlapping time of the segments), the vessel broadcast two or more normalized name, where each normalized name was bradcast at least 10 or more times. That is a bit complicated, but the goal is to identify overlapping segments where there were likely more than one identity. (this should be 0; if it is > 0, it can be used as a filter to remove potentially erroneous/noisy vessels)

We filter to just those vessels the random forest model calls fishing vessels (`random_forest_info.on_fishing_lis` and squid jiggers. A vessel is a squid jigger when `random_forest_info.best_vessel_class = squid_jigger`, or when the model rolls the vessel up to the generic `fishing` class but its finest-level prediction is `random_forest_info.atomic_class = squid_jigger` (GFW says this finest-level prediction is reliable for squid jiggers). Additionally, to reduce noise, we filter out vessels that broadcast exceedingly infrequently (i.e., `activity.active_hours < 24`) or are noisy vessels, using GFW's definition of a noisy vessel (i.e., `activity.offsetting` or `activity.overlap_hours_multinames >= 24`). They are simply not reliable and will not provide good effort estimates. This leaves us with 4,212 squid vessels globally.

1.5.1 Joined dataset: SST and AIS-based effort

One version of the final dataset we use for our analysis is a combination of the gridded AIS-based fishing effort data and gridded SST data. We start from every 0.25x0.25 degree pixel and month with SST data within the analysis scope, crossed with every flag that has fishing effort in the analysis scope. We then left join the AIS-based effort onto that grid by pixel, month, and flag, so pixel-months with no effort for a flag have zero effort. Each row in the joined dataset represents flag-level effort in a given pixel and month, with the corresponding SST for that pixel and month, the ONI for that month, and the pixel's EEZ information.

The joined dataset can be loaded in R using the command `targets::tar_load(joined_dataset_ais)`. The dataset has the following columns:

- **month**: Month (first day of month) (date)
- **lon_bin**: 0.25 degree longitude bin (degrees) (numeric)
- **lat_bin**: 0.25 degree latitude bin (degrees) (numeric)
- **flag**: Fishing vessel flag (ISO3) (character)
- **sst_deg_c_mean**: Mean sea surface temperature, aggregated from the raw daily 0.25x0.25 degree data (degrees C) (numeric). The dataset also has the standard deviation, minimum, and maximum (`sst_deg_c_sd`, `sst_deg_c_min`, `sst_deg_c_max`)
- **fishing_hours**: Total fishing effort across vessels (hours) (numeric)
- **fishing_kw_hours**: Total fishing effort across vessels (kW-hours) (numeric)
- **oceanic_nino_index**: Oceanic Niño Index for the month (degrees C) (numeric)
- **eez_id**, **eez_iso3**, **high_seas**: The Marine Regions v12 EEZ (MRGID) the pixel's centroid falls in, and its sovereign (ISO3), or `high_seas` (Institute (2023)). Joint regimes and disputed areas are not counted as EEZs, so pixels in them are `high_seas`
- **nearest_eez_id**, **nearest_eez_iso3**, **nearest_eez_distance_m**: The sovereign (Marine Regions sovereign MRGID and ISO3) of the pixel's EEZ, or for high seas pixels, of the nearest EEZ, and the distance (meters) from the pixel's centroid to the nearest 200 nautical mile boundary between an EEZ and the high seas

Here we summarize these data (Table 1.4):

Table 1.1: Summary statistics for joined dataset that includes gridded SST and AIS-based fishing effort

Table 1.1: Data summary

Name	dplyr::select(…)
Number of rows	37034760
Number of columns	7
_____	
Column type frequency:	
Date	1
character	1
numeric	5
_____	
Group variables	None

Variable type: Date

Table 1.2: Summary statistics for joined dataset that includes gridded SST and AIS-based fishing effort

skim_variable	n_missing	complete_rate	min	max	median	n_unique
month	0	1	2016-01-01	2025-12-01	2020-12-16	120

Variable type: character

Table 1.3: Summary statistics for joined dataset that includes gridded SST and AIS-based fishing effort

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
flag	5290680	0.86	3	3	0	6	0

Variable type: numeric

Table 1.4: Summary statistics for joined dataset that includes gridded SST and AIS-based fishing effort

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
lon_bin	0	1	-102.56	16.08	-130.0	-116.50	-102.75	-89.00	-70.25	
lat_bin	0	1	-15.81	14.44	-40.0	-28.25	-16.50	-3.25	10.00	
sst_deg_c_mean	0	1	22.81	4.19	10.3	19.92	23.51	26.24	31.11	
fishing_hours	0	1	0.11	7.31	0.0	0.00	0.00	0.00	4023.79	
fishing_kw_hours	0	1	108.74	7651.16	0.0	0.00	0.00	0.00	4038920.47	

1.6 Geographic analysis scope

Our proposed geographic scope encompasses a bounding box with a longitude range from -130 degrees to -70 degrees and a latitude range from -40 degrees to 10 degrees (Figure 1.1; Figure 1.2). This longitude range encompasses band of equatorial fishing effort to the west and the EEZs off the western coast of South America. The latitude range covers the maximum latitude of the South Pacific Regional Fisheries Management Organisation (SPRFMO) and

extends beyond the southern latitude where the north-south band of fishing effort is currently concentrated. The bounding box extends beyond where fishing effort is currently concentrated, which means that in our predictions under future climate change scenarios, we could capture shifts of fishing effort beyond it current range.

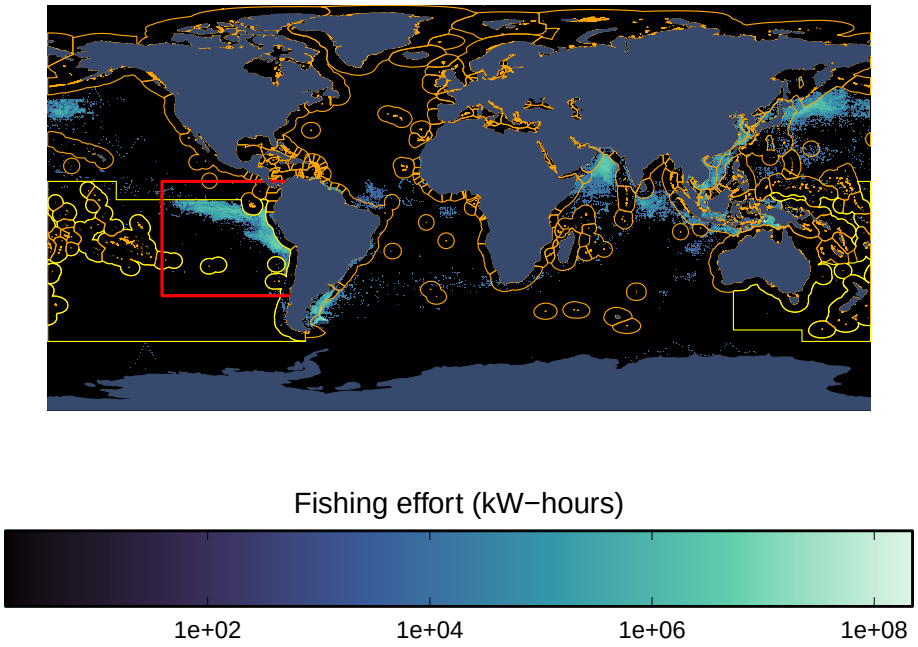


Figure 1.1: Map of squid jigger fishing effort from 2016 through December 2025, aggregating effort across effort, flags, and time for each pixel. EEZ boundaries from Marine Regions V12 are shown in orange; the SPRFRMO boundary is shown in yellow; the currently proposed analysis scope bounding box is shown as a red outline.

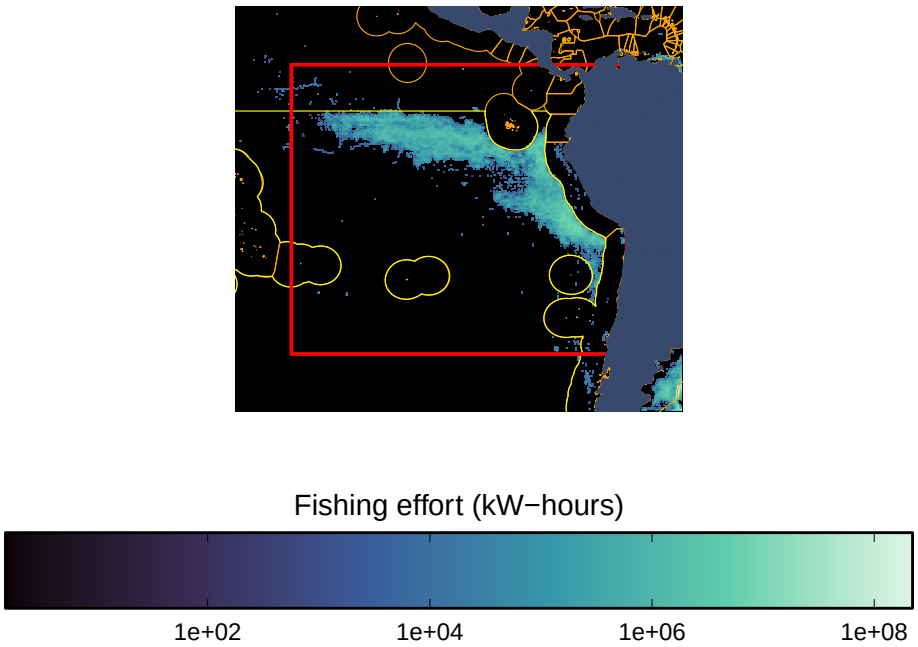


Figure 1.2: Zoomed in map of squid jigger fishing effort from 2016 through December 2025 to the proposed analysis scope, aggregating effort across effort, flags, and time for each pixel. EEZ boundaries from Marine Regions V12 are shown in orange; the SPRFRMO boundary is shown in yellow; the currently proposed analysis scope bounding box is shown as a red outline.

2 Exploratory data analysis

Note that all maps are zoomed into the area surrounded by the proposed geographic analysis scope.

2.1 Sea surface temperature (SST)

We can look at a map of SST, simply looking at the average monthly SST from across the entire January 2016 through December 2025 time series (Figure 2.1).

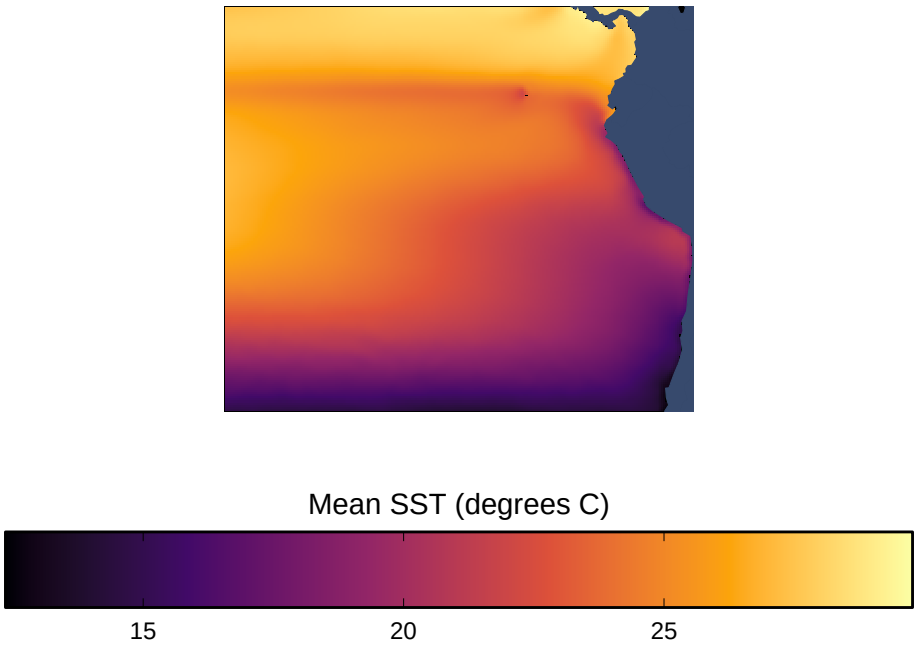


Figure 2.1: Map of mean sea surface temperature (SST) from across January 2016 through December 2025, using 0.25x0.25 degree pixels.

Aggregating across the mean sea surface temperatures of each pixel, we can calculate the mean sea surface temperature over time within our study scope (Figure 2.2). This allows us to see both seasonal trends and larger trends over time.

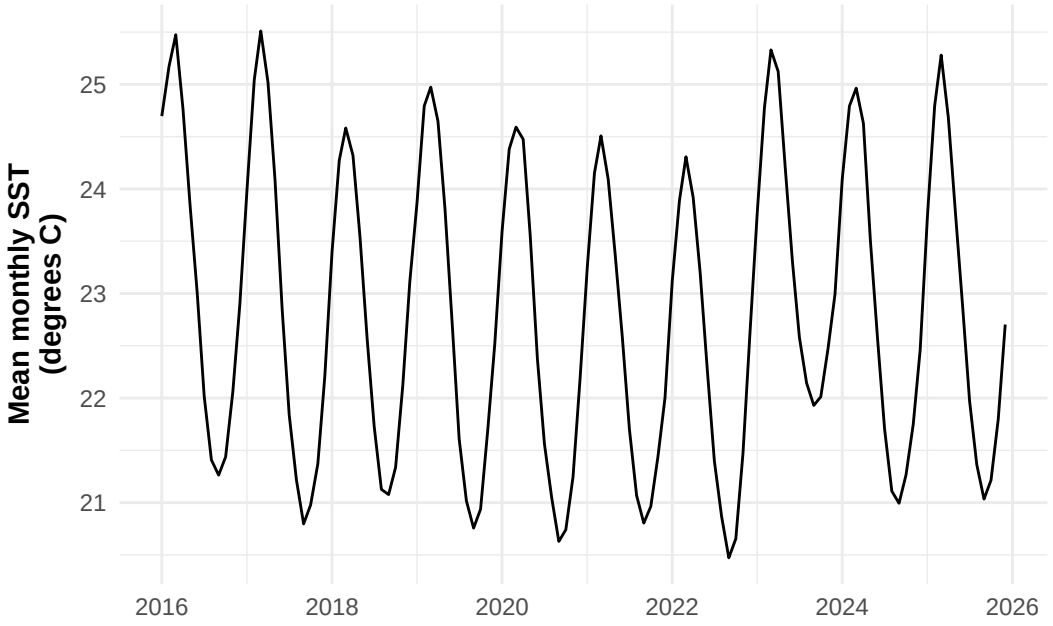


Figure 2.2: Time series of monthly mean sea surface temperature (SST) within our study scope.

2.2 Sea surface temperature (SST) forecasts under climate change

Here we look at a map of mean sea surface temperature (SST) under the four different climate change scenarios, and the three different forecast horizons, and focusing on the area of our analysis scope (Figure 2.3).

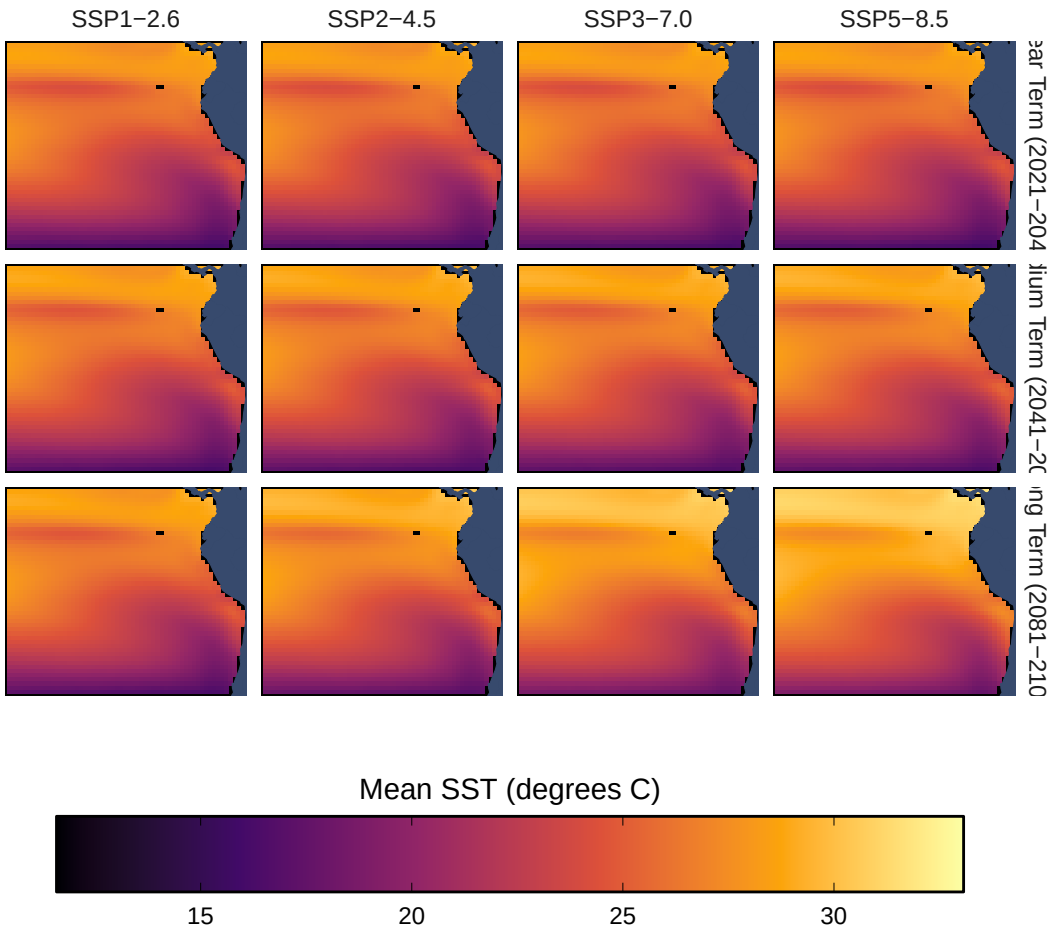


Figure 2.3: Map of mean August sea surface temperature (SST) under four different climate change scenarios, and three different forecast horizons. Data are for 1x1 degree pixels.

Here we look at time series of climate change forecasts for monthly average SST for each forecast time horizon and scenario, and focusing on our spatial analysis scope (Figure 2.4). As expected, projected SST is higher for time horizon further into the future, and for more extreme climate change scenarios.

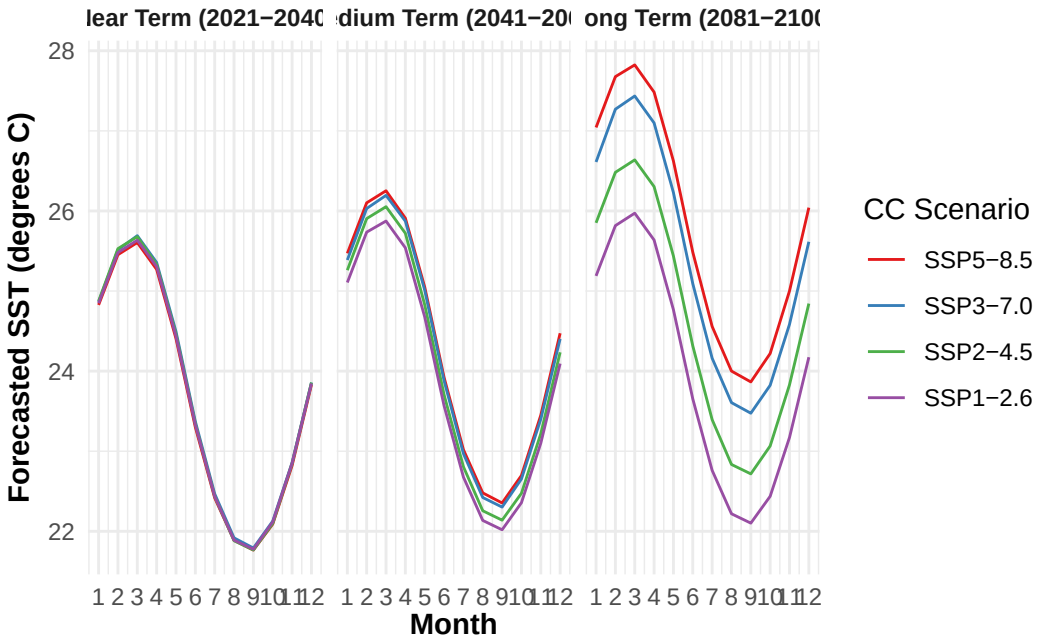


Figure 2.4: Climate change forecasts for monthly average SST for each forecast time horizon and scenario, within our spatial analysis scope

2.3 AIS-based Fishing effort

Here we look at total fishing effort from 2016 through December 2025 within the analysis scope, by flag (Figure 2.5). China has the most fishing effort, with 97.2% of all effort, followed by Taiwan with 0.8%.

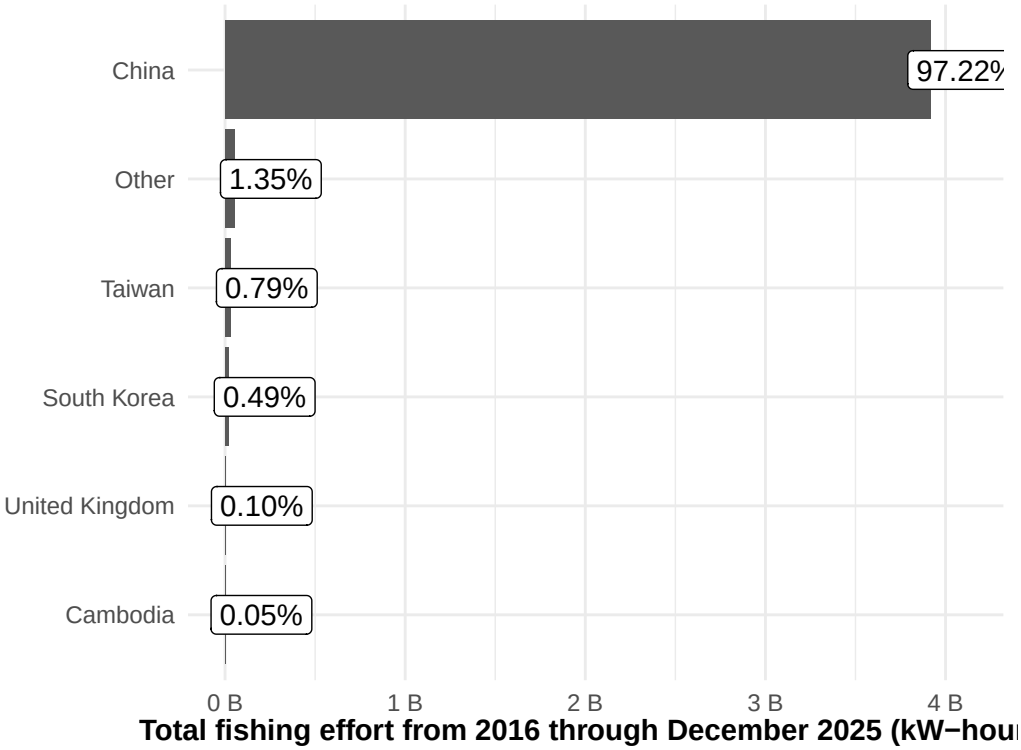


Figure 2.5: Total fishing effort, by flag, from 2016 through December 2025 within the analysis scope. The label shows the percentage of total fishing effort that each flag contributes. The top 5 flags are shown individually, with other flags aggregated into 'Other'.

Next we look at a time series of total monthly AIS-based fishing effort by fishing flag over time within the analysis scope (Figure 2.6). The top 5 flags are shown, with other flags aggregated into 'Other'.

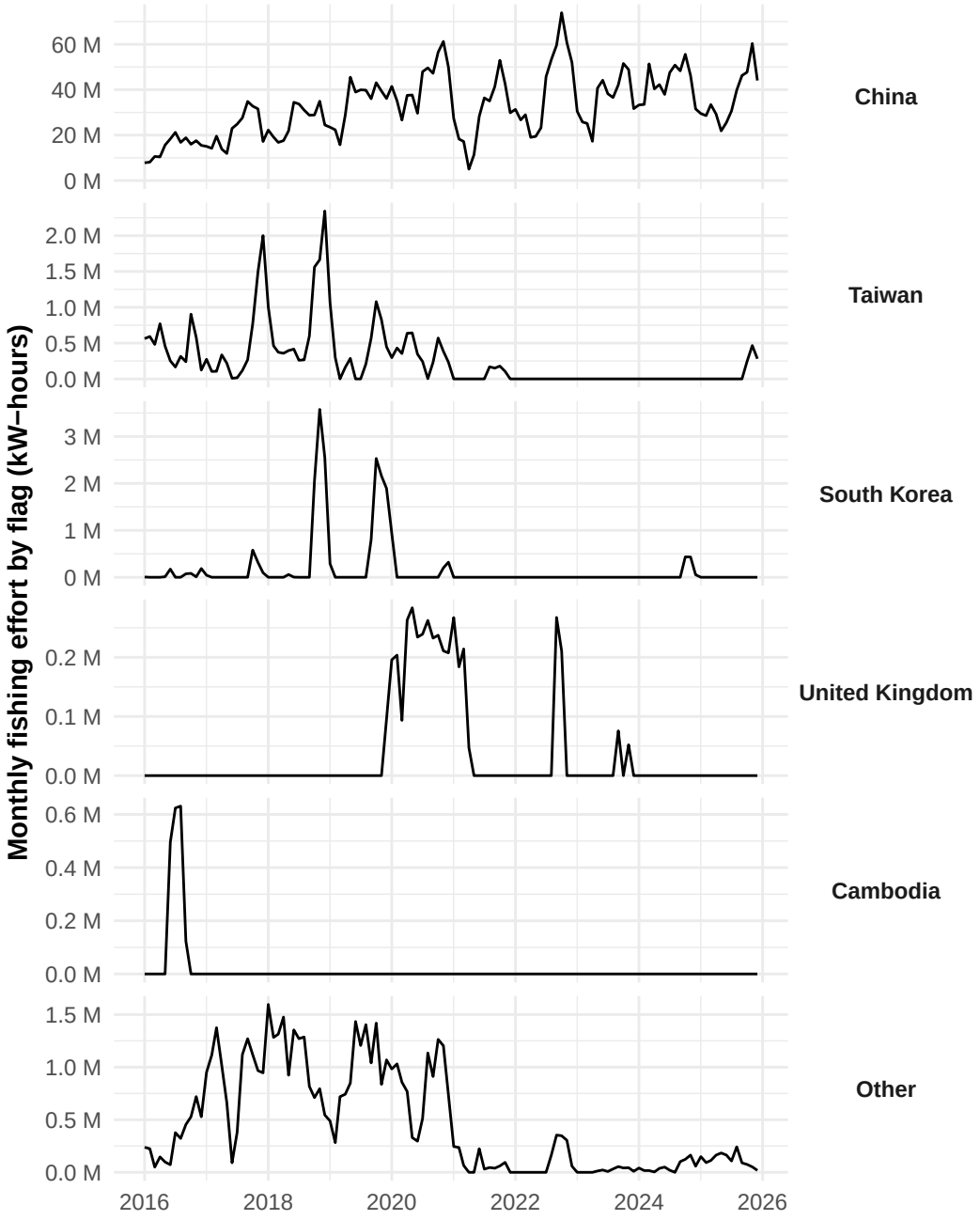


Figure 2.6: Monthly fishing effort by flag from 2016 through December 2025 within the analysis scope. The top 5 flags are shown individually, with other flags aggregated into 'Other'.

Next we look at a time series of total annual AIS-based fishing effort by fishing flag over time within the analysis scope (Figure 2.7). The top 5 flags are shown, with other flags aggregated into 'Other'.

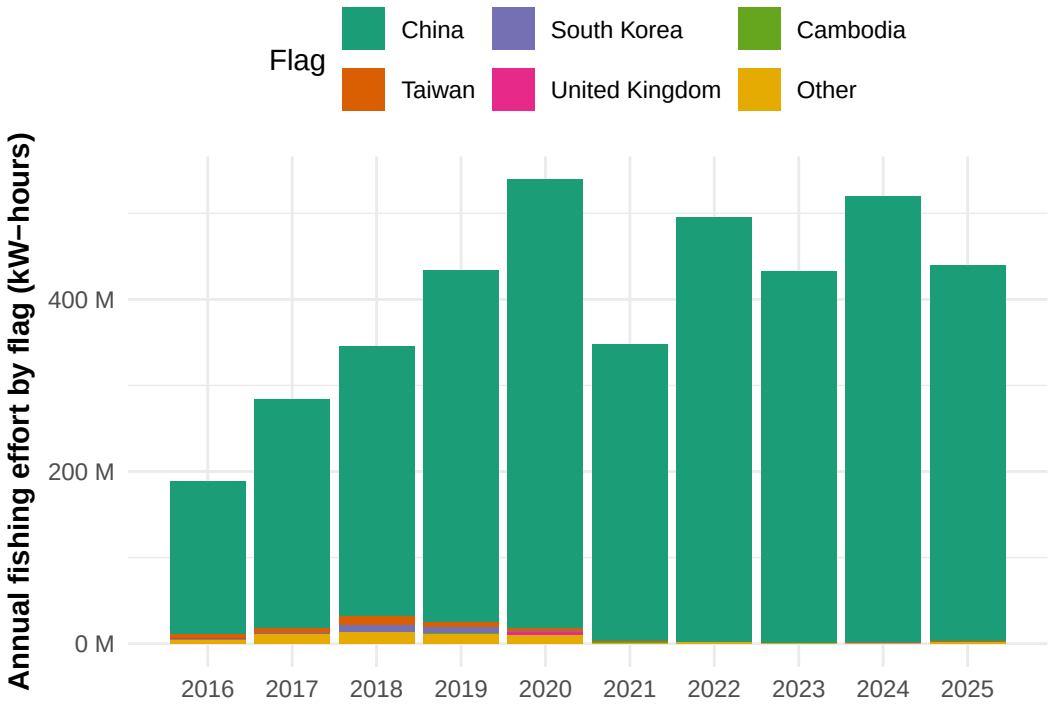


Figure 2.7: Annual fishing effort by flag from 2016 through December 2025 within the analysis scope. The top 5 flags are shown individually, with other flags aggregated into 'Other'.

Next we can look at the temporal trend of total fishing effort alongside the temporal trend of SST (Figure 2.8).

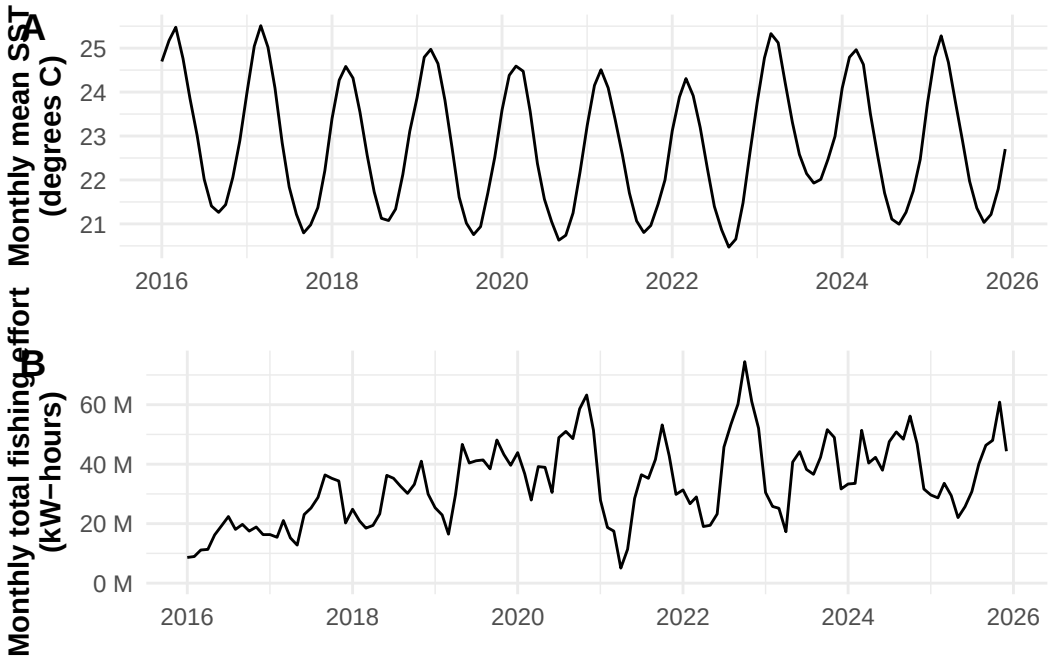


Figure 2.8: Monthly trends of of a) mean sea surface temperature (SST); and b) total squid jigger fishing effort. Both time series include data from across the entire January 2016 through December 2025 time period, and from only within our spatial analysis scope.

We can also aggregate the effort data by calculating the total effort for each month in each year, allowing us to look at the historic seasonal variation of total effort. We can do so for two regions: the "equatorial" region (latitude -10 to 10) and the "sub-equatorial" region (latitude -40 to -10) (Figure 2.9, Figure 2.10).

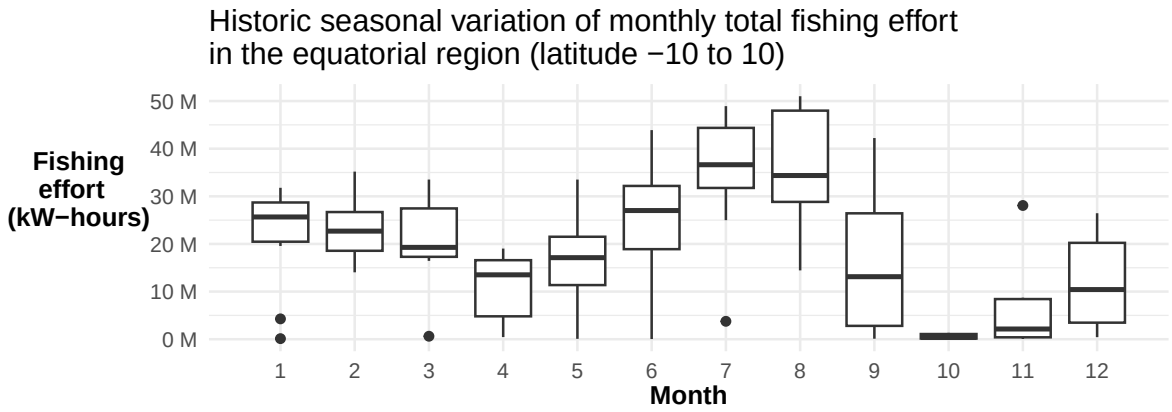


Figure 2.9: Historic seasonal variation of monthly total fishing effort for the equatorial region. The distribution for each month shows the spread of total fishing effort from each year in our historic dataset (January 2016 through December 2025) and within our spatial analysis scope.

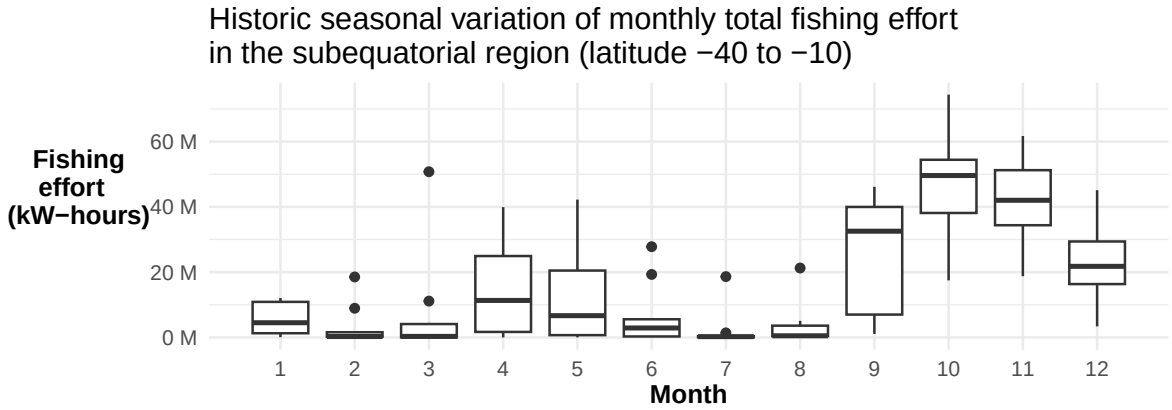


Figure 2.10: Historic seasonal variation of monthly total fishing effort for the sub-equatorial region. The distribution for each month shows the spread of total fishing effort from each year in our historic dataset (January 2016 through December 2025) and within our spatial analysis scope.

We can also aggregate the SST data by calculating the mean SST for each month in each year, allowing us to look at the historic seasonal variation of monthly mean sea surface temperature. We can do so for two regions: the "equatorial" region (latitude -10 to 10) and the "sub-equatorial" region (latitude -40 to -10) (Figure 2.11, Figure 2.12).

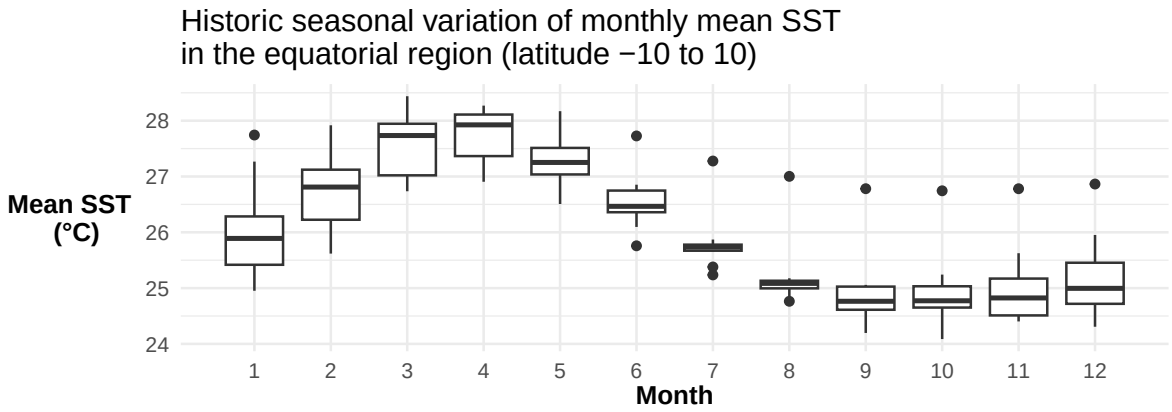


Figure 2.11: Historic seasonal variation of monthly mean sea surface temperature (SST) for the equatorial region. The distribution for each month shows the spread of mean SST from each year in our historic dataset (January 2016 through December 2025) and within our spatial analysis scope.

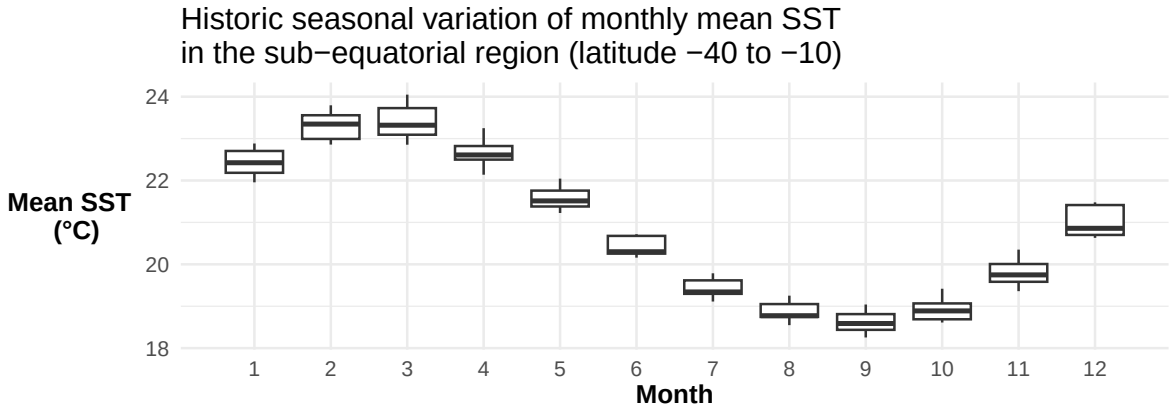


Figure 2.12: Historic seasonal variation of monthly mean sea surface temperature (SST) for the sub-equatorial region. The distribution for each month shows the spread of mean SST from each year in our historic dataset (January 2016 through December 2025) and within our spatial analysis scope.

Next we can look at at a map of AIS-based squid fishing effort (Figure 2.13), aggregating effort across effort and flags and time for each pixel across the entire processed time series. EEZ boundaries from Marine Regions v12 are shown in orange (Institute (2023)).

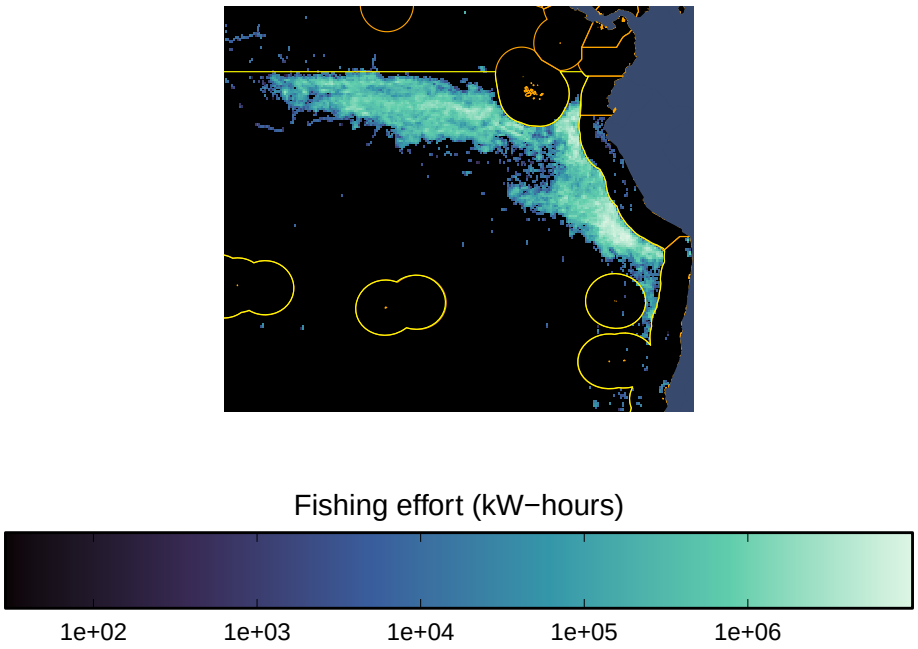


Figure 2.13: Map of squid jigger fishing effort from 2016 through December 2025, aggregating effort across effort, flags, and time for each pixel. EEZ boundaries from Marine Regions V12 are shown in orange; the SPRFMO boundary is shown in yellow.

We can also look at this map of total fishing effort alongside a map of mean SST (Figure 2.14).

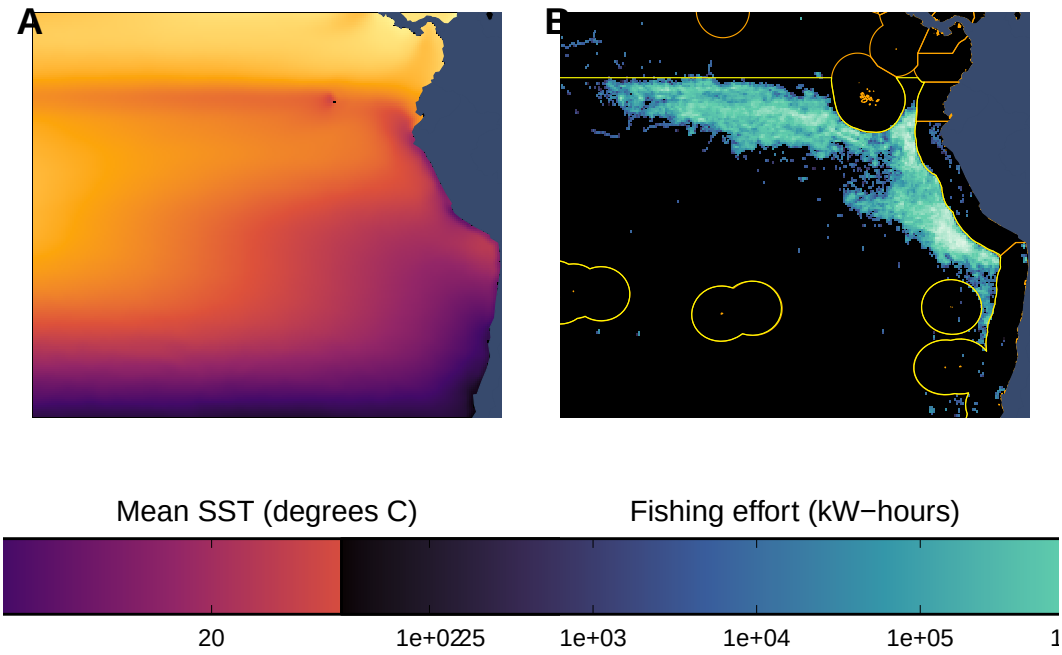
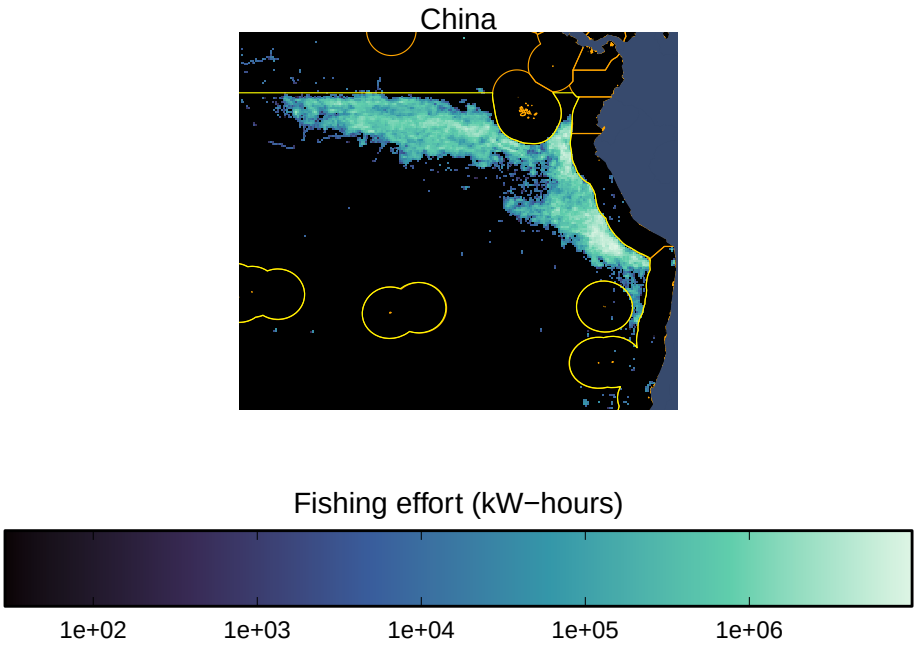


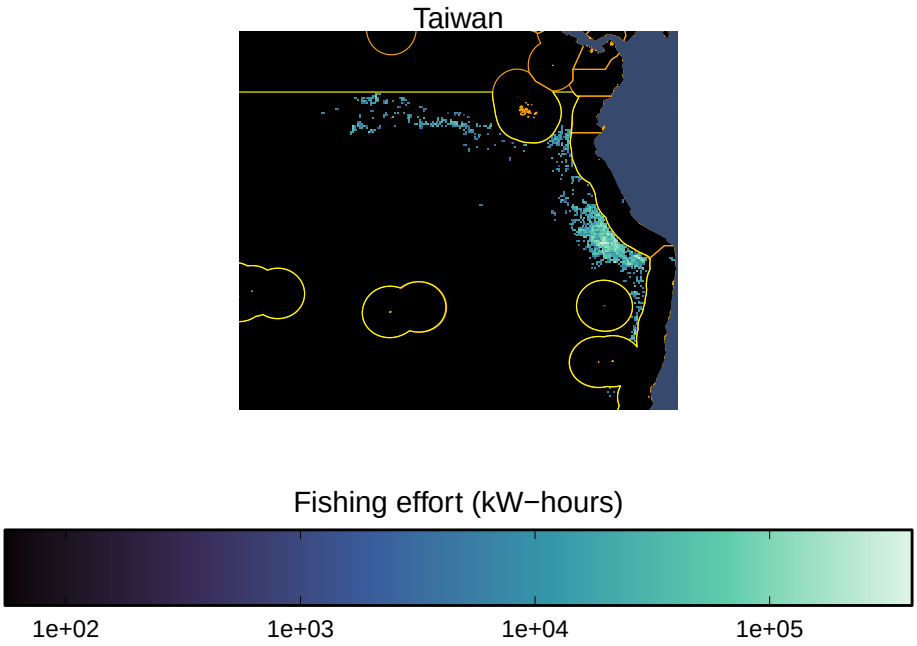
Figure 2.14: Maps of a) mean sea surface temperature (SST); and b) total squid jigger fishing effort (EEZ boundaries from Marine Regions V12 are shown in orange; the SPRFMO boundary is shown in yellow). Both maps include data from across the entire January 2016 through December 2025 time period, using 0.25x0.25 degree pixels.

We can also look at these effort maps, broken apart by flag, and still aggregating squid jigger fishing effort from 2016 through December 2025, aggregating effort across time for each pixel. The top 5 flags are shown.

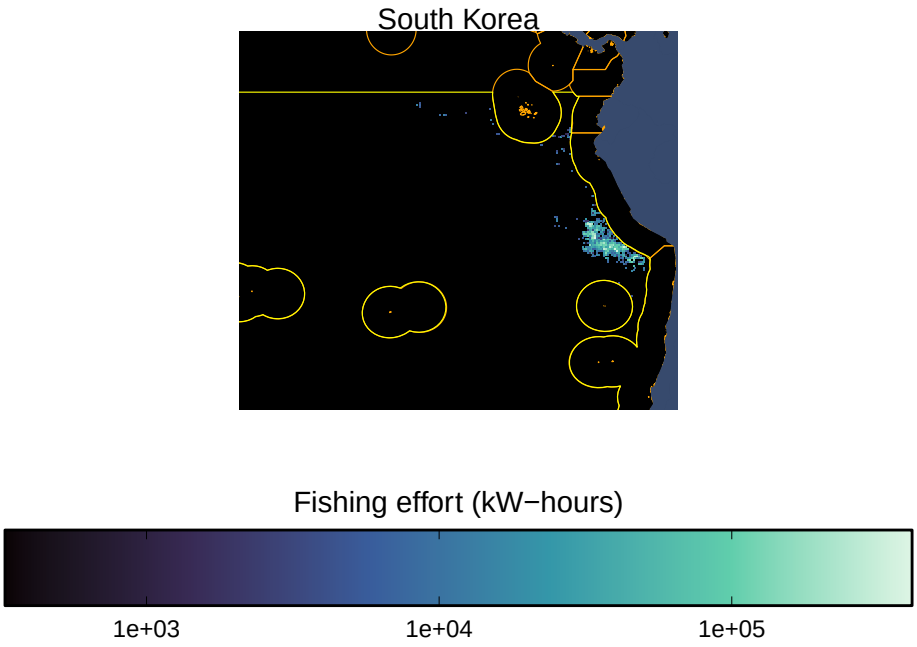
```
[[1]]
```


China
Fishing effort (kW−hours)
1e+02
1e+03
1e+04
1e+05
1e+06

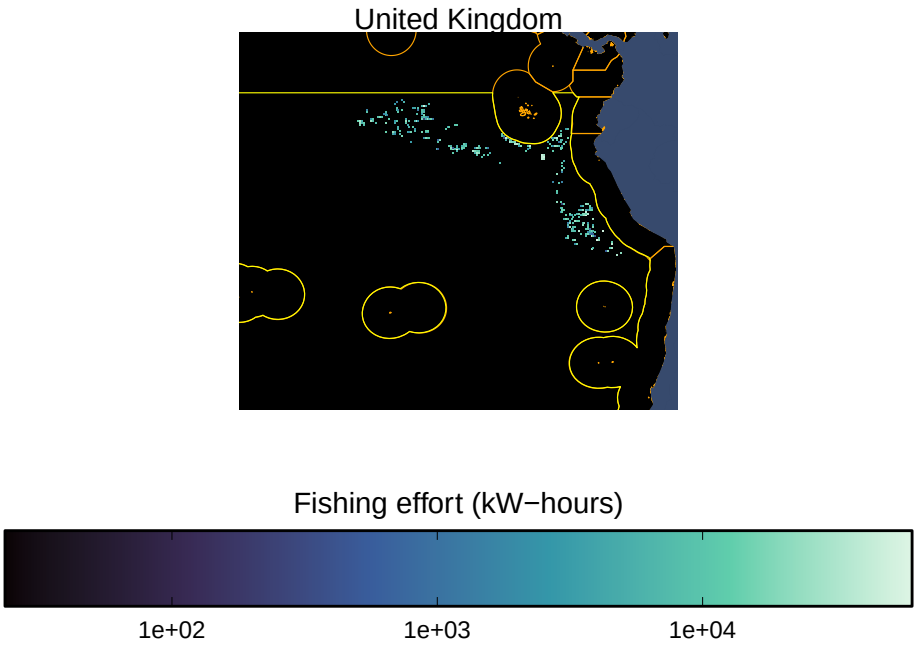
```
[[2]]
```


Taiwan
Fishing effort (kW−hours)
1e+02
1e+03
1e+04
1e+05

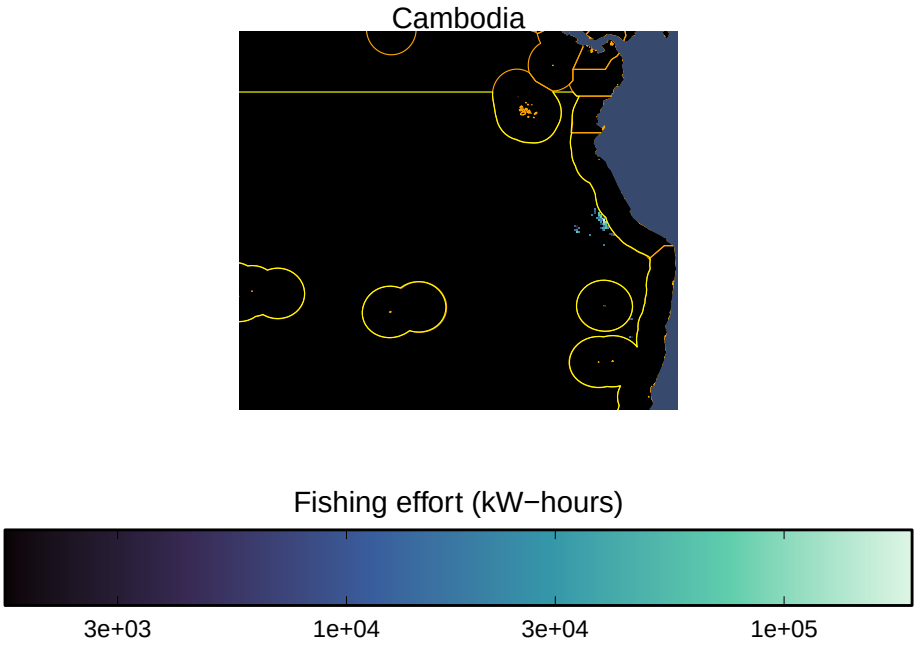
[[3]]



[[4]]


United Kingdom
Fishing effort (kW−hours)
1e+02
1e+03
1e+04

```
[[5]]
```


Cambodia
Fishing effort (kW−hours)
3e+03
1e+04
3e+04
1e+05

Change log

2026-11-02

What changed

AIS data. Squid jigger effort and vessel info now come from Global Fishing Watch's V5 pipeline instead of V3, and run from January 2016 through December 2025 (V3 ran through August 2024). The BigQuery tables are versioned (`_v20261102`), so the V3 tables are still there, unchanged. For 2016-2023, V5 has 13.4% more fishing hours and 9.6% more fishing kW-hours than V3 within the analysis scope; the difference is similar every year (9-17% more hours). The squid vessel list has 4,212 vessels.

Pipeline structure. The single `targets` pipeline is now three projects, run in order with `Rscript run.R`:

- `gfw` builds the AIS tables in BigQuery and exports them, and the VIIRS detections, to parquet files. It needs GFW BigQuery access.
- `sst` downloads daily SST, saves a dated copy of the ONI, and reads the CMIP6 forecasts. It needs internet.
- `analysis` joins everything and renders this book. It only reads files, so it can be rerun without GFW credentials or internet.

The README describes the safeguards: BigQuery tables are never overwritten, each query has a cost cap, the VIIRS tables are frozen, and downloads and joined datasets are checked before they're used.

SST. Daily SST now comes from NOAA's NCEI archive instead of the CoastWatch ERDDAP server. The values are identical for days both have, but ERDDAP was missing some days, and when a day was missing, it returned the nearest day instead. Four of the old daily files held the previous day's data (2021-03-23, 2021-09-07, 2022-10-19, and 2024-03-31), so those months counted one day twice. With NCEI, every day from 2016-01-01 through 2025-12-31 is included. Monthly SST is now only computed for pixels in the analysis scope.

ONI. The pipeline now uses a copy of NOAA's ONI file saved on 2026-11-02, and won't re-download it. NOAA revises past ONI values (it moved to ERSST v6 in September 2026): compared to the previous pull, values for 2017-2021 changed by up to 0.26°C, and 59 of those 60 months changed.

CMIP6 forecasts. The SST forecasts now come from the IPCC AR6 Interactive Atlas Dataset, the data behind the Interactive Atlas, instead of files downloaded by hand from the Atlas website. Of the 144 old files, 63 were copies of a different scenario or time horizon (e.g., Near Term SSP1-2.6 and SSP2-4.5 were identical for five months), the Near Term files had no February, and the Long Term files had two Februaries. The new forecasts follow the Atlas's own code for averaging across years and models, and reproduce the Atlas values to within 0.02°C. The pipeline also saves a 1995-2014 baseline from the historical runs, for each scenario's models (`sst_cc_baseline_data`), for computing projected changes.

EEZs. `nearest_eez_id` is now the Marine Regions sovereign ID for every pixel. Before, pixels in an EEZ had the EEZ's ID and high seas pixels had the sovereign's ID, so one country could have several IDs (e.g., Chile had four). The ISO3 columns didn't change.

Book. New ONI section; updated SST, CMIP6, and joined dataset descriptions; the scope maps in Chapter 1 now show effort outside the analysis scope; the annual effort figure is now a stacked bar chart through 2025; map captions now say 0.25x0.25 degree pixels; and figure cross-references are fixed.

Software. R 4.5.1, with packages updated to their latest versions (`renv.lock`), and Quarto 1.11.9.

Things to flag for Kaiwen

These are in Kaiwen's analysis files (`Analysis_forecast_ols.Rmd`, `Analysis_forecast_poisson.Rmd`, and `Visualization.Rmd`), which aren't part of the pipeline. Notes at the top of each file describe them. Only the data loading in those files changed: they now read the `analysis` project's store.

1. **Forecasts will change.** The CMIP6 files the forecasts used were mislabelled (see above), so the models' forecasts need to be rerun with the new `sst_cc_forecast_data`.
2. **CMIP6 is biased warm here.** The forecasts swap observed SST for absolute CMIP6 SST. Within the analysis scope, Near Term (2021-2040) CMIP6 SST averages 0.76°C warmer than observed SST for 2016-2025 (SSP1-2.6; 5-95% range of -0.48 to 2.67°C across 1x1 degree cells and months), and the 1995-2014 CMIP6 baseline is already 0.29°C warmer than observed 2016-2025 SST. So part of the predicted change is model bias rather than warming. The delta method (observed SST plus the CMIP6 change from the baseline, for the same cell and month) would avoid this; the baseline is now in the pipeline.
3. **Effort change maps understate effort about 7 times.** Rows are pixel x month x flag, and the models have no flag term, so every flag in a pixel-month gets the same prediction. Taking the mean by pixel and multiplying by 12 gives a per-flag number; there are 7 flags in scope.

4. **Wrong scenario.** In `Visualization.Rmd`, `ols_prediction_SSP1` is built from the SSP2-4.5 predictions, and the maps named `_ssp1` use SSP5-8.5.
5. **February and NAs.** With no Near Term February in the old files, February predictions were NA, and `mean(na.rm = TRUE)` and `replace_na(0)` showed February as no change. NA predictions should stop the script rather than be dropped.
6. **Region split.** `monthly_diff` splits the regions at `lat_bin <= -10`, but the models split at `lat_bin < -10`.
7. **Missing file.** `capture_quantity.csv` isn't in the repo or the project data folder.
8. **Threads.** The models use `nthreads = 20`, so run them in a Slurm job with at least 20 cores.
9. **`Visualization.Rmd` stops partway through.** A call to `plot_sst_change()` (the SST change map for January-June) leaves out the required `title` argument, so the script stops there; the chunk after it then can't find the plot. This was already in the code. Every other chunk ran.

Running Kaiwen's analysis files with the new data

On 2026-11-02, we ran the three files in order on sequoia (20 cores), with the new pipeline data. `Analysis_forecast_ols.Rmd` ran in 24 minutes and `Analysis_forecast_poisson.Rmd` in 19 minutes, both without errors. `Visualization.Rmd` stopped at the missing `title` argument (item 9); run with errors allowed, the rest of it ran. The forecasts in those runs still have the issues in items 2-6, so they aren't results yet.

References

Huang, Boyin, Chunying Liu, Viva Banzon, et al. 2021. "Improvements of the Daily Optimum Interpolation Sea Surface Temperature (DOISST) Version 2.1." *Journal of Climate* 34 (8): 2923–39.

Institute, Flanders Marine. 2023. "Maritime Boundaries Geodatabase, Version 12." In *Available Online at Https://Www.marineregions.org/. Https://Doi.org/10.14284/628*.

Iturbide, Maialen, Jesús Fernández, José Manuel Gutiérrez, et al. 2021. *Repository Supporting the Implementation of FAIR Principles in the IPCC-WG1 Atlas. Zenodo*. Intergovernmental Panel on Climate Change (IPCC) Geneva.

Kroodsma, David A, Juan Mayorga, Timothy Hochberg, et al. 2018. "Tracking the Global Footprint of Fisheries." *Science* 359 (6378): 904–8.

Park, Jaeyoon, Jennifer Van Osdel, Joanna Turner, et al. 2023. "Tracking Elusive and Shifting Identities of the Global Fishing Fleet." *Science Advances* 9 (3): eabp8200.